

Candidate's seat No.: _____

Charotar University of Science and Technology [CHARUSAT]
Faculty of Technology and Engineering
Department of Mathematical Sciences
MA243.01 TRANSFORMS AND VECTOR DIFFERENTIAL CALCULUS
1st UNIT TEST

Semester: 3rd Sem. B. Tech. (EC)

Maximum Marks: 30

Date: 05/08/2019 (Monday)

Time: 09:15 am to 10:15 am

Instructions:

- (i) Figures to the right indicate **full** marks.
- (ii) Use of scientific calculator is allowed.
- (iii) Draw figure where it is required.

Q-1 Choose the correct answer from the given options in the following: [06]

1. Which of the following function is the kernel of Laplace transform?
(a) $k(t, s) = se^{t^2}$ (b) $k(t, s) = e^{-st}$
(c) $k(t, s) = \sin st$ (d) $k(t, s) = \cos st$
2. The Laplace transform of $t^{\frac{1}{2}}$ is _____.
(a) $\frac{\pi}{2s^{\frac{3}{2}}}$ (b) $\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$
(c) $\frac{\pi}{s^{\frac{3}{2}}}$ (d) $\frac{\sqrt{\pi}}{s^{\frac{3}{2}}}$
3. The value of convolution of 1 and 1 is _____.
(a) 0 (b) t^2
(c) t (d) $-t$
4. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then the magnitude of the vector $r^2\vec{r}$ is _____.
(a) $(x^2 + y^2 + z^2)^{\frac{3}{2}}$ (b) $(x^2 + y^2 + z^2)^2$
(c) $(x^2 + y^2 + z^2)^3$ (d) $x^2 + y^2 + z^2$
5. Which of the following vector is unit vector of $\vec{a} = (1, -1, 1)$?
(a) $\frac{1}{2}(1, -1, 1)$ (b) $\frac{1}{\sqrt{2}}(1, -1, 1)$
(c) $\frac{1}{\sqrt{3}}(1, -1, 1)$ (d) $\frac{1}{3}(1, -1, 1)$
6. The rotation of a vector in $\vec{F}$ is in clockwise direction, if _____.
(a) $\nabla \times \vec{F} \neq 0$ (b) $\nabla \times \vec{F} = 0$
(c) $\nabla \times \vec{F} > 0$ (d) $\nabla \times \vec{F} < 0$

Q-2 Attempt any Three.**[12]**

- (a) I. Prove that $L\{e^{at}\} = \frac{1}{s-a}$ for all $s > a$.
- II. Prove that $L\{\cos at\} = \frac{s}{s^2+a^2}$ for all $s > 0$.
- (b) Define conservative field. Find the divergence and curl of the vector $f(x, y, z) = x^2y \hat{i} + xz \hat{j} + 2yz \hat{k}$ at point $(-1, 1, 1)$. Find the value of $\nabla \cdot (\nabla \times f)$.
- (c) Using convolution theorem, find the inverse Laplace transform of
$$\frac{s^2}{(s^2 + a^2)(s^2 + b^2)}.$$
- (d) Find the inverse Laplace transform of
$$\frac{4s + 5}{(s - 1)^2(s + 2)}.$$

Q-3 Attempt any Three.**[12]**

- (a) Find the unit tangent and unit normal vector at $t = 2$ along the curve $x(t) = t^2 - 1, y(t) = 4t - 3, z(t) = 2t^2 - 6t$.
- (b) Find the directional derivative of a scalar function $\phi(x, y, z) = xy^2 + yz^3$ at the point $(1, -1, 1)$, (i) along the direction $\hat{i} + 2\hat{j} + 2\hat{k}$ and (ii) towards the point $(2, -1, 1)$.
- (c) Find Laplace transform of $\frac{1-\cos t}{t}$ and $e^{-2t} \sin 3t$.
- (d) Show that $\bar{V} = 2xyz \hat{i} + (x^2z + 2y) \hat{j} + (x^2y) \hat{k}$ is the irrotational and find a scalar potential function $\phi(x, y, z)$ such that $\bar{V} = \text{grad}(\phi)$.
